

# Access-profile detail

Selected-return states are separate from the historical scores in Figures 2–4.

| Task                                           |                                     | Selected return |       | Archived M3 detail |       | Archived M4 version |       |
|------------------------------------------------|-------------------------------------|-----------------|-------|--------------------|-------|---------------------|-------|
|                                                |                                     | CANN            | CUDA‡ | CANN               | CUDA‡ | CANN                | CUDA‡ |
| Environment and installation                   |                                     |                 |       |                    |       |                     |       |
| I                                              | Version compatibility               | S               | S     | 4                  | 5     | 4                   | 4     |
| J                                              | Installation                        | S               | S     | 5                  | 5     | 2                   | 3     |
| K                                              | Container setup                     | S               | S     | 5                  | 4     | 4                   | 5     |
| Operator development                           |                                     |                 |       |                    |       |                     |       |
| C                                              | Library selection                   | S               | S     | 3                  | 5     | 3                   | 5     |
| D                                              | Custom operator                     | N               | S     | —                  | 5     | 2                   | 3     |
| L                                              | Operator accuracy                   | S               | S     | 5                  | 5     | 2                   | 5     |
| M                                              | Dynamic shapes / tiling             | S               | S     | 4                  | 5     | 4                   | 4     |
| N                                              | Operator fusion                     | S               | S     | 5                  | 4     | 2                   | 5     |
| O                                              | Framework integration               | S               | S     | 5                  | 5     | 3                   | 3     |
| Training                                       |                                     |                 |       |                    |       |                     |       |
| F                                              | Distributed training                | S               | R     | 4                  | 5     | 2                   | 3     |
| P                                              | Mixed precision                     | S               | S     | 4                  | 5     | 4                   | 5     |
| Q                                              | Memory / OOM                        | S               | S     | 5                  | 5     | 4                   | 5     |
| R                                              | Training convergence                | S               | S     | 5                  | 4     | 3                   | 5     |
| Inference and deployment                       |                                     |                 |       |                    |       |                     |       |
| A                                              | Model conversion                    | S               | S     | 4                  | 5     | 2                   | 4     |
| H                                              | Quantization                        | S               | R     | 3                  | 5     | 2                   | 3     |
| S                                              | Inference serving                   | S               | S     | 4                  | 5     | 3                   | 4     |
| T                                              | Dynamic inference                   | S               | S     | 5                  | 5     | 2                   | 5     |
| Performance and optimization                   |                                     |                 |       |                    |       |                     |       |
| B                                              | Profiling                           | S               | S     | 4                  | 5     | 2                   | 4     |
| U                                              | Memory / occupancy                  | S               | S     | 4                  | 5     | 5                   | 5     |
| V                                              | Compute-transfer overlap            | S               | S     | 4                  | 5     | 2                   | 5     |
| W                                              | Multi-device communication          | S               | S     | 3                  | 4     | 3                   | 4     |
| Debugging                                      |                                     |                 |       |                    |       |                     |       |
| E                                              | Error-code diagnosis                | S               | R     | 2                  | 5     | 2                   | 5     |
| X                                              | Memory-error diagnosis              | S               | S     | 5                  | 5     | 2                   | 3     |
| Migration                                      |                                     |                 |       |                    |       |                     |       |
| Y                                              | Chip migration                      | S               | S     | 4                  | 5     | 3                   | 4     |
| Z                                              | Programming concepts                | S               | S     | 5                  | 4     | 5                   | 5     |
| Migration analogy (not in CANN/CUDA summaries) |                                     |                 |       |                    |       |                     |       |
| G                                              | Migration analogy [CANN / ROCm-HIP] | S               | R     | 4                  | 5     | 4                   | 3     |

S = substantive text; N = navigation only; R = no selected-read reassessment. S does not imply all task requirements are met.

‡ For G, CUDA‡ = ROCm/HIP; it is a migration analogy and is excluded from CANN/CUDA summaries.